

EVAN M. PLATTEN

585-739-6880 | evan_platten@urmc.rochester.edu

EDUCATION

University of Rochester <i>Master of Science in Data Science</i> Overall GPA: 3.97	Rochester, NY May 2026
Stony Brook University <i>Bachelor of Science in Biochemistry, Magna Cum Laude</i> Overall GPA: 3.73	Stony Brook, NY May 2019
Monroe Community College (SUNY) <i>Associate in Science, Liberal Arts & Sciences: General Studies</i> Overall GPA: 3.34	Rochester, NY 2017

HONORS

- Stony Brook University: Dean's List, 4 of 4 semesters

RESEARCH EXPERIENCE

University of Rochester – Department of Biochemistry & Biophysics <i>Research Specialist I – Dr. Eric Phizicky</i> - Performed RNA biology research with a focus on tRNA biology. - Expanded technical repertoire to include advanced RNA analysis techniques: Northern blot analysis for transcript-level detection and quantification, and mass spectrometry analysis for biomolecular characterization. - Independently applied existing bioinformatics, programming (Python, R), and data-analysis skills to support ongoing research questions in the lab.	Rochester, NY March 2025 – Present
<i>Research Specialist I – Dr. Elizabeth Grayhack</i> - Uncovered novel links between the Integrated Stress Response pathway, implicated in cognitive disorders and neurodegeneration, and pathways involved in the maintenance of protein expression. - Discovered synergistic interactions between Integrated Stress Response protein Gcn1, implicated in neurodegeneration, and Translational Quality Control components. - Designed and constructed genetic tools to isolate mutant yeast strains that suppress specific aberrant protein-expression phenotypes. - Analyzed whole-genome sequencing data using custom Python pipelines. - Designed GFP/RFP fluorescent fusion-construct reporter assays and ran them on the LSRII flow cytometer (URMC Flow Core certified).	July 2019 – March 2025

PROJECTS

Spatial Transcriptomics Educational Viewer <i>DSC 483 Capstone & DSC 495 Master's Research</i> - Lead developer (Unity 3D, C#, Python). Designed and built an interactive Unity application for visualizing and teaching spatial transcriptomics data. - Architected the application around a cross-scene event system enabling dataset hot-swapping, modular UI components, and a complete guided tutorial flow with state-driven progression between scenes. - Implemented high-performance point-cloud rendering of Visium HD spatial-transcriptomics datasets. - Wrote a Python export pipeline converting AnnData objects into flat files usable by Unity.	Aug 2025 – May 2026
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- Resolved cross-platform build issues for both WebGL and standalone targets.
- Authored a comprehensive transition guide documenting the full system architecture for future work.

Yeast Strain Database — Grayhack / Phizicky Laboratories

DSCC 461 Database Systems — Graduate Course Team Project | SQL, PHP, HTML

- Course project framed as a corporate-level design-and-build, requiring the team to define requirements, design the schema, and implement a working database from scratch with no provided starter materials.
- Designed a relational schema in SQL with normalized tables, foreign-key relationships, and indexed lookup fields to track yeast strain inventory, genotypes, parentage, and experimental metadata for the labs.
- Built a PHP/HTML web interface for laboratory users to search, filter, and update strain records without writing SQL — a working example of a stakeholder-facing data product for non-technical scientists.

PUBLICATIONS

Bruno, B.S., **Platten, E.M.**, Houston, L., Brule, C.E., Grayhack, E.J. (2026). The Tor pathway, ribosome concentration, and wobble decoding mediate inhibitory effects of the Leu-Pro CUC-CCG codon pair in *Saccharomyces cerevisiae*. *bioRxiv* 2026.02.20.707122. <https://doi.org/10.64898/2026.02.20.707122>

Houston, L., **Platten, E.M.**, Connelly, S.M., Wang, J., Grayhack, E.J. (2022). Frameshifting at collided ribosomes is modulated by elongation factor eEF3 and by Integrated Stress Response regulators Gcn1 and Gcn20. *RNA*, 28(3): 320–339. PMID: PMC8848926.

TEACHING EXPERIENCE

Stony Brook University — Department of Physics

Stony Brook, NY

PHY 475 — Undergraduate Teaching Practicum

- Served as an undergraduate teaching assistant in a physics course, supporting instruction and student learning.

Stony Brook University — Department of Biology

Stony Brook, NY

BIO 475 — Teaching Practicum, College Biology I

- Served as an undergraduate teaching assistant in College Biology I, supporting instruction and student learning.

TECHNICAL SKILLS

- **Programming:** Python (NumPy, Pandas, Requests, BioPython, scikit-learn, Matplotlib) — data analysis, bioinformatics scripting, ML model implementation.
- **Statistical Computing:** R and RMarkdown — statistical analysis, hypothesis testing, regression, reproducible reports.
- **Databases:** SQL — relational schema design, querying, indexing; exposure to NoSQL concepts.
- **Machine Learning:** Supervised and unsupervised learning, dimensionality reduction (PCA, t-SNE), SVMs, decision trees, ensemble methods, neural networks, model evaluation and selection.
- **Bioinformatics Tools:** Whole-genome sequencing analysis, BLAST, IGV (Integrative Genomics Viewer), sequence trace viewers (e.g., Chromas), BioPython, AnnData / spatial transcriptomics.
- **Wet-Lab:** Yeast genetics (*S. cerevisiae*) — strain construction, plasmid design, recombinant DNA, Gibson assembly, ligation cloning; PCR and agarose gel purification; RNA isolation, DNase treatment, reverse transcription, RT-qPCR; Northern blot analysis; mass spectrometry analysis; flow cytometry (LSRII, certified by URMCI Flowcore).
- **Visualization & Software Engineering:** Unity 3D and C# — interactive 3D applications, point-cloud rendering, WebGL/standalone builds.
- **Other:** Advanced Excel formula development for whole-genome sequencing analysis; self-taught Linux user with experience administering virtual machines and lab computing infrastructure; Git-based development.

LABORATORY OPERATIONS & MENTORSHIP

University of Rochester – Department of Biochemistry & Biophysics

Dr. Eric Phizicky & Dr. Elizabeth Grayhack

- Trained and supervised undergraduate researchers and incoming technicians in laboratory methods, data analysis, experimental design, and lab safety.
- Trained co-workers in advanced Excel use for whole-genome sequencing analysis, and in IGV interpretation of WGS read maps.
- Managed lab supply ordering, inventory, and invoice reconciliation with the University finance office.
- Oversaw lab IT — workstation maintenance, troubleshooting, data-storage setup, and coordination with university IT department on ongoing projects.
- Maintained lab equipment, conducted weekly eyewash testing, and coordinated hazardous waste removal with University waste management.
- Notified and coordinated with appropriate offices for shared-equipment failures and scheduled maintenance.

SERVICE & VOLUNTEER EXPERIENCE

Friends of Nevada Wilderness — AmeriCorps

Reno, NV

Volunteer (300-hour service term)

- Completed a 300-hour AmeriCorps service term contributing to wilderness conservation and stewardship in northern Nevada.